

Multiple commenters stated that many broker-dealers, particularly those with retail customers, would have to incur additional costs for investor education and customer assistance in order to handle any investor confusion arising from the amendments to Rule 612.¹⁶⁶⁴ One commenter specifically mentioned that the amendments to Rule 612 would complicate “good-till-cancelled” orders (“GTC orders”) as the orders could be placed under one tick size and could still be active after a tick size change.¹⁶⁶⁵ This scenario is unlikely given that there will be a period of one-month between end of the evaluation period and the tick size implementation during which market participants will be able to know which stocks will be changing their tick size. For an issue to arise, the GTC order would have to be in place over the course of that month and the trader would have to be unaware of the upcoming tick size change. Retail facing broker-dealers will likely implement some method of notifying their customers that the tick size for some stocks will change following the end of the evaluation period. To the extent to which customer confusion causes these costs to materialize, the Commission would expect that these costs would likely be greater in the time immediately following the implementation of the amendments, and they would decrease over time as investors become accustomed to the new tick size regime.

b. Estimates for Rule 610

As in the Proposing Release, the Commission estimates a \$57,000 one-time cost to exchanges to comply with changes to Rule 610.¹⁶⁶⁶ This estimate assumes that exchanges will combine in the same Rule 19b-4 filing their proposals to amend their fees and rebates and make

¹⁶⁶⁴ See, e.g., FISD Letter at 2, TastyTrade Letter at 19, and TradeStation Letter at 6.

¹⁶⁶⁵ See TradeStation Letter at 6.

¹⁶⁶⁶ See Proposing Release, supra note 11, at 80333.

fees and rebates determinable at the time of execution, and that this process will not increase the cost of those filings. The Commission recognizes that if these filings are not efficiently combined, then the costs to exchanges could be higher than \$57,000. The Commission estimates also assume that LTSE will not file a proposed rule change with the Commission because it does not currently charge access fees or offer rebates, but that the other 15 exchanges will file proposed rule changes. If so, these amendments will lead to an estimated one-time total cost of \$855,000 across exchanges to comply with Rule 610.¹⁶⁶⁷

c. Estimates for Rules 600 and 603

The exclusive SIPs and exchanges will incur compliance costs associated with accelerating the inclusion of odd-lot data inside the NBBO in SIP data, adding the BOLO to SIP data, and accelerating the implementation of the round lot definitions as amended in this release. The round lot definitions (but for amendments to them in this release) and the inclusion of odd-lot data inside the NBBO were both parts of the MDI Rules. Thus, the amendments will accelerate the compliance costs associated with these aspects of the MDI Rules. One difference is that the MDI Rules anticipated that these changes to NMS data would occur after the competing consolidator model was up and running. Thus, the MDI Rules did not anticipate that the exclusive SIPs would incur such costs unless they chose to become competing consolidators. The addition of the best odd-lot order to the SIP data was not part of the MDI Rules and will thus be a new cost under the amendments. Accordingly, the discussion below distinguishes costs to

¹⁶⁶⁷ The Commission does not expect other market participants to incur significant incremental costs associated with the change in the access fees and rebates. As shown in Table 4, market participants deal with over 100 fee changes per year across all exchanges and thus it is reasonable to expect that one fee change by the exchanges to bring their fees into compliance with these amendments would represent an economically trivial incremental cost to these market participants.

the exclusive SIPs in terms of those included in the MDI Rules and new costs from these amendments.

The Commission estimates a one-time cost of \$3,500 and ongoing costs of \$6,500 per year for at least two years for exchanges to comply with the amendments to Rules 603 and 600.¹⁶⁶⁸ This estimate accounts for the acceleration of the necessary data to generate the odd-lot information, including the best odd-lot order, and transmit this information to the exclusive SIPs. The costs reported here account for an increase in the costs associated with the MDI Rules that will require the exchanges to transmit to competing consolidators all of the data necessary for generating consolidated market data.

Consequently, for the exchanges, the costs associated with providing the exclusive SIPs with odd-lot information will represent an acceleration of costs anticipated in the MDI Rules rather than new costs, with a few differences. First, the odd-lot information will be transmitted to the exclusive SIPs as opposed to the competing consolidators. Second, the ongoing costs of these amendments will be incurred only until the exclusive SIPs are retired, which the Commission estimates will be at least two years after the Commission's approval of the plan amendment(s) required by Rule 614(e).

Compliance with the amendments to Rules 603 and 600 will require the exclusive SIPs to develop, operate, and maintain systems to collect and disseminate the odd-lot information inside the NBBO as well as the best odd-lot order. The Commission expects that these costs will

¹⁶⁶⁸ In the MDI Adopting Release, supra note 10, at 18764, the Commission estimated costs to the exchanges of collecting and transmitting the necessary information to the competing consolidators to be approximately \$70,000 in one-time costs and approximately \$130,000 in ongoing costs. The additional \$3,500 in one-time costs and \$6,500 in ongoing costs represent a 5% addition over the costs in the MDI release to account for the new requirement to send the necessary data to generate odd-lot information to the exclusive SIPs (i.e., $\$3,500 \approx \$70,000 \times 0.05$ and $\$6,500 \approx \$130,000 \times 0.05$). See infra note 1841 and accompanying text.

primarily consist of costs that an exclusive SIP would incur if it were to convert to a competing consolidator. Thus, for exclusive SIPs that would have become competing consolidators in the absence of these amendments, initial compliance costs represent an acceleration of costs under the MDI Rules, rather than new additional costs. Further, the ongoing costs for exclusive SIPs to comply with Rules 600 and 603 will be incurred only until the exclusive SIPs are retired, after which time these costs will consist of ongoing costs that were previously accounted for in the MDI Rules. The Commission estimates that the exclusive SIPs will incur a one-time cost of approximately \$613,000 and ongoing costs of approximately \$174,000 per year.¹⁶⁶⁹

The Commission recognizes some uncertainty in the assumption that exclusive SIPs will become competing consolidators. If one or both exclusive SIPs are not planning to become competing consolidators under the MDI Rules, and the amendments do not change their plans, then the estimated initial and ongoing costs in Table 15 represent new costs associated with the amendments. If the amendments were to prompt one or both exclusive SIPs to become competing consolidators, when they otherwise would not have done so, then the costs in Table 15 underestimate the total costs of these SIPs becoming competing consolidators. In the MDI Rules, however, the Commission anticipated that both exchanges operating exclusive SIPs would

¹⁶⁶⁹ The \$613,000 estimate in one-time costs is based on the following calculations: \$33,000 (costs under the amendments to Rule 612 to update data specifications and internally and externally test the updates; see supra note 1651, see also section VII.D.5.a) + \$167,670 (\$83,790 (hourly rate for Sr. Programmer at \$399 for 210 hours) + \$61,740 (hourly rate for Sr. Systems Analyst at \$343 for 180 hours) + \$7,460 (hourly rate for Compliance Manager at \$373 for 20 hours) + \$5,880 (hourly rate for Director of Compliance at \$588 for 10 hours) + \$8,800 (hourly rate for Compliance Attorney at \$440 for 20 hours) + \$412,500 (costs for external services). The \$174,000 estimate in ongoing costs is based on the following calculations: \$50,301 (\$25,137 (hourly rate for Sr. Programmer at \$399 for 63 hours) + \$18,522 (hourly rate for Sr. Systems Analyst at \$343 for 54 hours) + \$2,238 (hourly rate for Compliance Manager at \$373 for 6 hours) + \$1,764 (hourly rate for Director of Compliance at \$588 for 3 hours) + \$2,640 (hourly rate for Compliance Attorney at \$440 for 6 hours)) + \$123,725 (costs for external services). See infra notes 1830, 1832, 1834 and 1835 and accompanying text for relevant details on these cost estimates.

have strong incentives to enter the competing consolidator market.¹⁶⁷⁰ The amendments require that the exclusive SIPs build out the capacity to disseminate aspects of the data required by the MDI Rules. This could increase the likelihood that the exclusive SIPs will choose to become competing consolidators because they will already have implemented some of the technology needed to comply with the requirements of a competing consolidator, thereby lowering their subsequent cost of becoming a competing consolidator. In this context, the Commission continues to expect that the exclusive SIPs will become competing consolidators, and the estimated costs in Table 15 largely represent costs that the exclusive SIPs would have borne in the process of becoming competing consolidators.

The Commission recognizes that the amendment to Rule 600 could increase the initial costs of becoming a competing consolidator as well as the ongoing costs of competing consolidators, but the Commission believes that such costs are already accounted for in the MDI Rules.¹⁶⁷¹ In particular, competing consolidators could incur additional compliance costs to estimate and disseminate the best odd-lot order. To the extent such costs are not accounted for in the MDI Rules, they will likely be a small fraction of the compliance costs of including odd-lot information in SIP data stated above. Indeed, the competing consolidators will already have the information necessary to calculate the BOLO, so most of the cost incurred under the amendment to Rule 600 will be the initial cost of coding the information and the cost of processing and monitoring that code in real time.

¹⁶⁷⁰ See MDI Adopting Release, supra note 10, at 18761.

¹⁶⁷¹ See supra section VII.D.5 for further discussion of how or whether this requirement would alter the compliance costs of competing consolidators.

6. Interactions with Recently Adopted Rules

The Commission acknowledges that the effects of any final rule may be impacted by recently adopted rules that precede it. Accordingly, each economic analysis in each adopting release considers an updated economic baseline that incorporates any new regulatory requirements, including compliance costs, at the time of each adoption, and considers the incremental new benefits and incremental new costs over those already resulting from the preceding rules. We discuss below economic effects stemming from interactions between the final rule and other recently adopted rules.¹⁶⁷²

a. Amendments to Rule 605

Commenters have specifically questioned the Commission's analysis of interactions between the four EMS Proposals,¹⁶⁷³ of which only the Rule 605 Proposal has been adopted.¹⁶⁷⁴ Because the amendments to Rule 605 were not yet adopted at the time of the Proposing Release and were thus not a part of the baseline in the Proposing Release,¹⁶⁷⁵ the economic effects described in the Proposing Release may differ from those described here to the extent those effects change due to the amendments to Rule 605. Below, we discuss specific impacts the amendments to Rule 605, which is now part of the baseline, may have on the expected economic effects of the final rules compared to description of those effects in the Proposing Release, as well as the impact the final rules may have on the effects of amended Rule 605. In response to comments, we also consider whether the amended Rule 605 data is needed to assess the impact

¹⁶⁷² As explained above, the Commission considers recently adopted rules, but not recent proposals, as part of its baseline against which it measures the economic effects of its rules. See supra section VII.C and notes 1034 and 1047.

¹⁶⁷³ See supra section II for further discussion.

¹⁶⁷⁴ See Rule 605 Amendments, supra note 10.

¹⁶⁷⁵ See supra note 1672.

of the final rules. Overall, the inclusion of the amendments to Rule 605 in the baseline does not significantly change the costs and benefits of the final rules, and the final rules adopted herein have significant benefits even taking into account the adopted amendments to Rule 605 as part of the baseline.

i. Impact of Amended Rule 605 on the Final Rules

First, the Commission considers whether the amendments to Rule 605, which are now part of the baseline, may have affected certain benefits of the final rules compared to how those benefits were described in the Proposing Release. Some commenters stated that the four EMS Proposals, including the amendments to Rule 605 and these final rules, have similar objectives, such that the benefits of each rule may be overlapping, and that therefore each successive rule would have fewer benefits than were described in each proposing release.¹⁶⁷⁶ However, the final rules address different and significant issues in the national market system distinct from those addressed in Rule 605.¹⁶⁷⁷ The final rules have benefits, such as certain improvements in market quality for stocks that receive a smaller tick size, lower trading costs for liquidity demanders in certain stocks that experience a reduction in their access fees, and increased transparency and reduced complexity of exchange access fees and rebates, that are distinct from the benefits

¹⁶⁷⁶ See, e.g., Virtu Letter II at 55-56 (stating “that the proposals are designed to accomplish the same overarching goals,” that “each rule ignores the possibility that the other three rules may already address the Commission’s concerns,” that “the expected benefit the Commission believes its rules will achieve is overlapping,” and that an important policymaking question that arises from these overlapping objectives is whether “the estimated benefits [are] purely additive”) see also SIFMA Letter II at 100 (stating that “the Proposed Rules may, each individually, largely affect the same aspects of equity markets, including the economics of liquidity provision, spreads (particularly for retail investors), and costs (particularly for wholesalers)”; Virtu Letter II at 20 (stating that, if any of the other rules (including Rule 605) are successful at achieving their stated purpose, “competition would be enhanced without the Proposed [Tick Size] Rule (and its significant risks and costs) and the claimed benefits of the Proposed Rule are overstated”); Virtu Letter III at 2 (stating that the amendments in the Rule 605 Amendments may “otherwise address any concerns that formed the impetus for the [EMS] Proposals,” including the Proposing Release).

¹⁶⁷⁷ See supra section II for further discussion.

resulting from the amendments to Rule 605 and could not conceivably have been achieved through the amendments to Rule 605.¹⁶⁷⁸ While, as one commenter stated,¹⁶⁷⁹ both rules may improve competition, the issues being addressed in these final rules and in the amendments to Rule 605, and the mechanisms used to address them, differ significantly, making these benefits additive rather than overlapping. For example, the amendments to Rule 605 will increase competition among trading venues through greater transparency,¹⁶⁸⁰ while these final rules will increase competition between orders on trading venues in some stocks by removing barriers to subpenny quoting. Both of these competitive effects are expected to improve execution quality, but through different mechanisms and independently of one another.

In addition, the adoption of the amendments to Rule 605 may, to some extent, enhance certain benefits of the final rules compared to the benefits described in the Proposing Release. Specifically, the final rules are expected to increase the extent to which broker-dealers make decisions based on execution quality.¹⁶⁸¹ At the same time, the amendments to Rule 605 improve broker-dealers' access to information about the execution quality of market centers.¹⁶⁸² Thus, to the extent that broker-dealers base their order routing decisions more on execution quality as a

¹⁶⁷⁸ See supra sections VII.D.1.b, VII.D.2, and VII.D.3 for further discussion of the benefits of the final rules and supra section VII.C.5 for a discussion of the benefits resulting from the amendments to Rule 605.

¹⁶⁷⁹ See supra note 1676.

¹⁶⁸⁰ See Rule 605 Amendments, supra note 10, at 26543-75.

¹⁶⁸¹ Specifically, the increase in transparency in exchange access fees and rebates in the final rules is expected to decrease the extent to which broker-dealers' routing decisions are based on access fees and rebates and increase the extent to which these decisions are based on other factors, including the execution quality of market centers See supra section VII.D.3.

¹⁶⁸² For example, the amendments to Rule 605 increase the granularity of time-to-execution buckets, which will improve broker-dealers' ability to compare execution speeds across trading venues and route their orders accordingly. See Rule 605 Amendments, supra note 10, at 26561.

result of the final rules, the improved access to market center execution quality information under amended Rule 605 will help facilitate those decisions.¹⁶⁸³

The adopted amendments to Rule 605 may also, in certain circumstances, cause the benefits of the final rules stemming from transparency to be somewhat lower than those described in the Proposing Release; however, the Commission expects these impacts to be minor. Specifically, in the Rule 605 Amendments, the Commission anticipated that the increase in transparency and competition on the basis of execution quality as a result of the amendments to Rule 605 might make broker-dealers less likely to route customer orders based on exchange fees and rebates.¹⁶⁸⁴ At the same time, the amendments to Rule 610 that would make fees and rebates determinable at the time of execution are expected to reduce broker-dealer conflicts of interest related to fees and rebates.¹⁶⁸⁵ Likewise, the lower access fee cap is also expected to reduce broker-dealer conflicts of interest.¹⁶⁸⁶ If the amendments to Rule 605 result in exchange fees and rebates becoming less important for broker-dealer customers' order routing decisions, the benefits resulting from a reduction in conflicts of interest under the final rules—caused by reducing the access fee cap and increasing the transparency of exchange fees and rebates—may be reduced compared to how they were described in the Proposing Release. However, this reduction in benefits, compared to the Proposing Release, is likely to be minor, for several reasons, and the Commission still expects the benefits described above, in comparison to the

¹⁶⁸³ See Rule 605 Amendments, supra note 10, at 26544-26547 (discussing the impact of the amendments to Rule 605 on competition between broker-dealers).

¹⁶⁸⁴ See Rule 605 Amendments, supra note 10, at 26586.

¹⁶⁸⁵ See supra section VII.D.3 for a discussion of the economic effects of requiring exchange fees and rebates to be determinable at the time of execution.

¹⁶⁸⁶ See supra section VII.D.2 for a discussion of the economics effects of reducing the access fee cap on conflicts of interest.

baseline, to be realized. First, not all orders are subject to and directly benefit from increased transparency under amended Rule 605.¹⁶⁸⁷ Second, the amended Rule 605 reporting requirements only require reporting by larger broker-dealers; while these broker-dealers handle the vast majority of customer accounts, they only handle around 60% of customer order flow in terms of number of orders.¹⁶⁸⁸ Therefore, the amendments to Rule 605 are not expected to directly impact customer order routing decisions for a significant subset of order flow,¹⁶⁸⁹ such that the final rules lowering the access fee cap and increasing the transparency of exchange fees and rebates are still expected to have additional benefits above and beyond those of the amendments to Rule 605.

Third, the Commission anticipated that the amendments to Rule 605, by expanding the scope of covered orders to include odd-lots, would encourage broker-dealers to compete for these orders on the basis of execution quality.¹⁶⁹⁰ The final rule's requirement that the exclusive SIPs disseminate odd-lot data¹⁶⁹¹ is expected to accelerate the benefits of accelerating the implementation of including odd-lot information in NMS data,¹⁶⁹² which may include facilitating

¹⁶⁸⁷ For example, in the Rule 605 Amendments, an analysis of Tick Size Pilot data found that, between April 2016 and March 2019, approximately 25% of orders were flagged as having special handling requests, which would exclude them from the scope of Rule 605 reporting requirements. See Rule 605 Amendments, supra note 10, at 26514.

¹⁶⁸⁸ The Rule 605 Amendments estimated that only 85 out of 1,245 broker-dealers with at least one customer account would qualify as a larger broker-dealer and therefore be required to prepare Rule 605 reports; however, these 85 broker-dealers are responsible for more than 98% of customer accounts and more than 60% of customer orders. See Rule 605 Amendments, supra note 10, at 26428 (Table 13).

¹⁶⁸⁹ The Rule 605 Amendments acknowledge that, if smaller broker-dealers are also incentivized to produce execution quality information for their customers as a result of the expanded scope of Rule 605 to include larger broker-dealers, the benefits of increased competition could indirectly extend to smaller broker-dealers as well. See Rule 605 Amendments, section IX.C.1.(D)(1), supra note 10, at 26428.

¹⁶⁹⁰ See Rule 605 Amendments, supra note 10, at 26552.

¹⁶⁹¹ See 17 CFR 242.603(b)(3) for rule text relating to this requirement under Rule 603.

¹⁶⁹² See supra section VII.D.4.c.